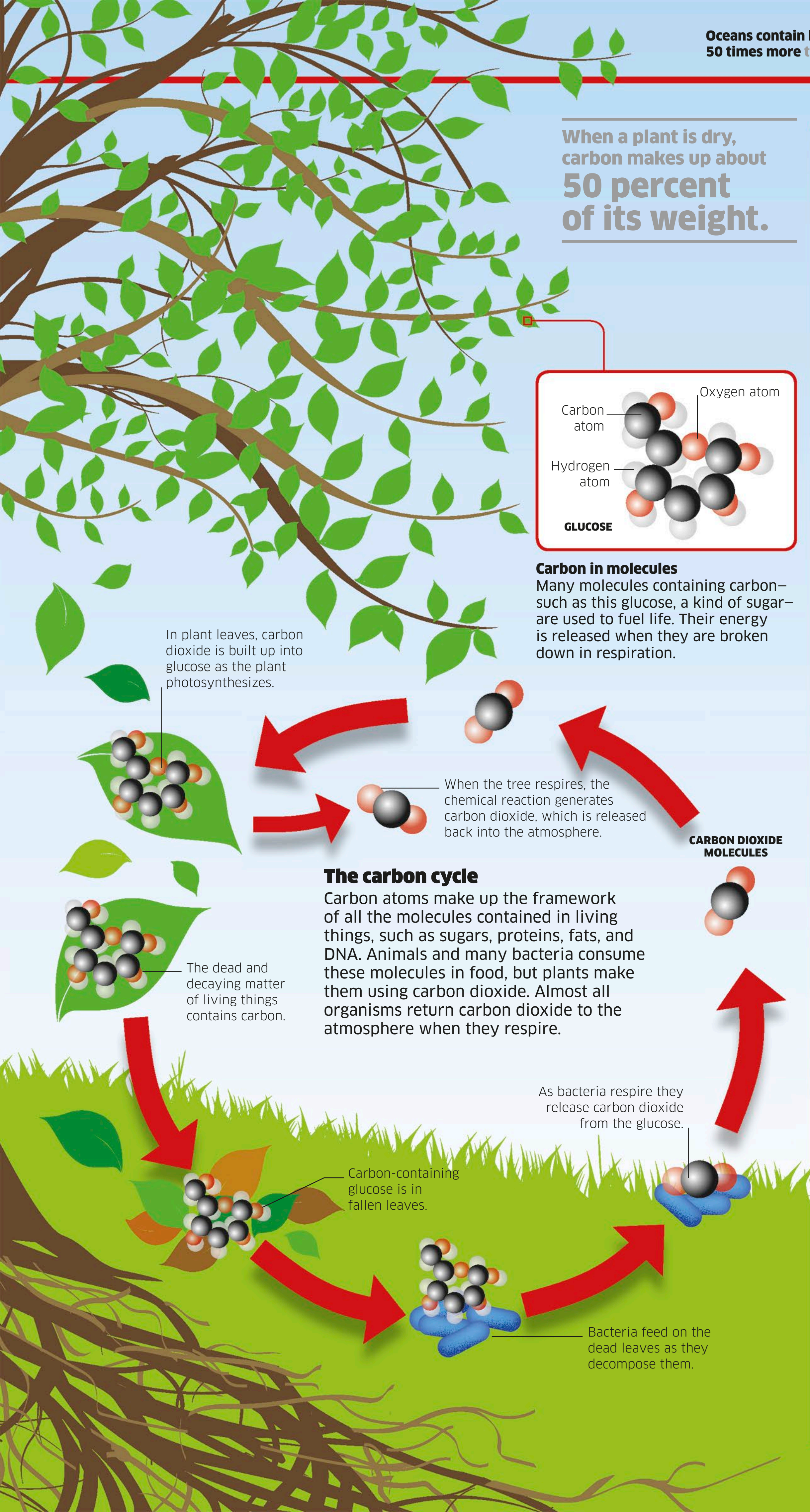
When a plant is dry, carbon makes up about **50 percent of its weight.**

Carbon in molecules
Many molecules containing carbon—such as this glucose, a kind of sugar—are used to fuel life. Their energy is released when they are broken down in respiration.

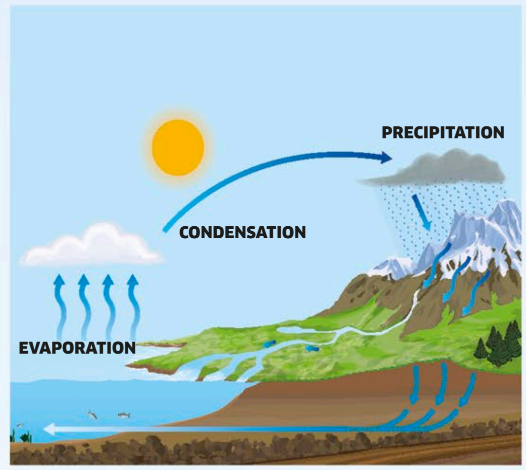


The carbon cycle

Carbon atoms make up the framework of all the molecules contained in living things, such as sugars, proteins, fats, and DNA. Animals and many bacteria consume these molecules in food, but plants make them using carbon dioxide. Almost all organisms return carbon dioxide to the atmosphere when they respire.

Recycling water

Water is made of two elements—hydrogen and oxygen—and travels through earth, sea, and sky in the global water cycle. This cycle is dominated by two processes: evaporation and precipitation. Liquid water in oceans, lakes, and even on plant leaves evaporates to form gaseous water vapor. The water vapor then condenses to form the tiny droplets inside clouds, before falling back down to Earth as precipitation: rain, hail, or snow.



The water cycle
Recycling of water is driven by the heating effects of the sun. At it is warmed, surface water evaporates into the atmosphere, but cools and condenses to form rain or snow. Rainfall drains or runs off to oceans and lakes to complete the cycle.

The long-term carbon cycle

Carbon atoms can be recycled between living organisms and the air within days, but other changes deeper in the earth take place over millions of years. Lots of carbon gets trapped within the bodies of dead organisms either in the ocean or underground—forming fossil fuels. It is then only released back into the atmosphere through natural events such as volcanic eruptions, or when it is burned by humans in forms such as coal (see pp.36–37).

Coal mining
At this mining terminal in Australia, carbon-containing coal is extracted from the ground.